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SECRETARY OF THE AIR FORCE

DEPARTMENT OF THE AIR FORCE
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Cyberspace Operations

SPECTRUM MANAGEMENT

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This publication implements Department of the Air Force Policy Directive 17-2, *Cyber Warfare Operations* and is consistent with the guidance established with National Telecommunications and Information Administration (NTIA), *Manual of Regulations and Procedures for Federal Radio Frequency Management* (also known as "Redbook") and the Joint Staff J6 Military Command, Control, Communications, and Computers Executive Board (MC4EB) Frequency Panel. It identifies various levels of responsibilities for Department of the Air Force spectrum management and use of the electromagnetic spectrum. This publication applies to all civilian employees and uniformed members of Regular Air Force, the Air Force Reserve, the Air National Guard, the United States Space Force, those who are contractually obligated to comply with Department of the Air Force publications, and the Civil Air Patrol when conducting missions as the official Air Force Auxiliary. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program,* and are disposed in accordance with the Air Force Records Disposition Schedule which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility using the Air Force Form 847, *Recommendation for Change of Publication;* route Air Force Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the office of primary responsibility of this publication for coordination prior to certification and approval. The authorities to waive wing unit-level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Instruction 33-360, *Publications and Forms Management*, for a description of the authorities

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SUMMARY OF CHANGES

This publication has been substantially revised and needs be completely reviewed. Significant reductions have been made and several chapters and process-oriented areas eliminated. Guidance for those chapters is available in other publications (e.g., NTIA Manual, Department of Defense Instructions (DoDI), Department of Defense Directives, and Allied Communications Publications (ACP)).

1. OVERVIEW.

1.1. The electromagnetic spectrum is the range of frequencies of electromagnetic radiation extending from zero to infinity, measured in hertz (cycles per second). The radio frequency spectrum is the group of frequency sub-bands within the allocated electromagnetic spectrum associated with radio wave propagation between 9 kilohertz and 275 gigahertz. This instruction provides for the Department of the Air Force's responsibilities in the conduct of peacetime electromagnetic spectrum management.

1.2. International law, domestic law, and implementing regulations require effective management and use of the electromagnetic spectrum. Within the United States, the electromagnetic spectrum is allocated between federal and non-federal users with portions of the spectrum being shared as noted in the NTIA Manual national table of allocations. The Department of Commerce's NTIA Manual regulates federal operations and the Federal Communications Commission regulates non-federal operations. Federal users must utilize frequency bands allocated for government or shared use in accordance with NTIA Manual. **(T-0)** A government frequency assignment is authorized in a non-government allocated frequency band, providing the Federal Communications Commission approves the request.

1.3. Sovereign nations exercise control over the use of the electromagnetic spectrum within their own territory. This basic consideration of international spectrum management becomes extremely important when United States military forces operate abroad. In nearly all circumstances, it is necessary to obtain host nation approval to conduct electromagnetic spectrum operations before United States military forces operate any spectrum-dependent system in a foreign nation. Misuse of spectrum resources or unauthorized electromagnetic spectrum use within a host nation may be a violation of international treaties, international law, or national laws and regulations. Seek assistance from the servicing staff judge advocate with understanding international treaties, regulations, and host nation laws, where applicable.

2. ROLES AND RESPONSIBILITIES.

2.1. Department of the Air Force Spectrum Operations.

2.1.1. Deputy Chief of Staff, Intelligence, Surveillance, Reconnaissance, and Cyber Effects Operations (AF/A2/6). In accordance with the Department of the Air Force Policy Directive 17-2, AF/A2/6 provides policy, guidance, resource advocacy and oversight for Air Force electromagnetic spectrum management in support of the Department of the Air Force mission and retains responsibility for policy, as well as oversight of lead command implementation of DoDI 4650.01 and DoDI 8320.05.

2.1.2. Director, Cyberspace Operations and Warfighter Communications (AF/A2/6C). AF/A2/6C is responsible for integrating cyberspace warfare capabilities into the joint mission and provides oversight and control for electromagnetic spectrum management in support of the Department of the Air Force mission.

2.1.3. Chief, Department of the Air Force Spectrum Management (AF/A2/6CO). AF/A2/6CO represents the United States Air Force and United States Space Force electromagnetic spectrum use at the national and international levels. Processes Department of the Air Force spectrum supportability and frequency assignment requests through the NTIA for use within the United States and its possessions. The AF/A2/6CO represents the United States Space Force, as a service, at the MC4EB Frequency Panel until the United States Space Force has established its spectrum offices. AF/A2/6CO will:

2.1.3.1. Implement national, international, and joint staff electromagnetic spectrum management policy in accordance with the NTIA Manual; International Telecommunication Union, *Radio Regulations* (ITU RR), DoD, Joint Staff, and MC4EB publications. **(T-0)**

2.1.3.2. Evaluate Department of the Air Force plans for needed electromagnetic support per DoDI 4650.01, *Policy and Procedures for Management and Use of the Electromagnetic Spectrum*. **(T-0)**

2.1.3.3. Represent and defend Department of the Air Force electromagnetic technical interests in national-level committees, groups, and organizations that address electromagnetic spectrum management matters in accordance with DoDI 4650.01. **(T-0)**

2.1.3.4. Negotiates at the departmental and national levels to obtain spectrum certification, host nation coordination through combatant commands, frequency assignments, electromagnetic attack training authorizations. Participates, as an invited member, of the US delegation led by the State Department and NTIA for Satellite Bilateral coordination and the World Radiocommunication Conference for the International Telecommunication Union (ITU) radio regulation changes. Obtains ITU satellite registration to satisfy the Department of the Air Force's orbital slots and radio frequency requirements for use in the electromagnetic spectrum in accordance with the NTIA Manual and ITU RR. **(T-0)**

2.1.3.5. Provide spectrum guidance to Department of the Air Force sponsored Department of Defense area frequency coordinators in accordance with ACP 190 US SUPP-1(D), *Guide to Frequency Planning*. **(T-0)**

2.1.3.6. Assist in resolution of interference problems involving the Department of the Air Force's assigned frequencies, per AFI 17-221, *Spectrum Interference Resolution Program.*

2.1.3.7. Provide guidance through appropriate command channels on electromagnetic spectrum to developers and users of all Department of the Air Force systems that require electromagnetic spectrum access or whose performance is influenced by radio frequency energy. This includes communications and information systems, electromagnetic warfare operations, test and training, intelligence and weapons systems, commercial-off-the-shelf equipment, and any other spectrum-dependent systems that rely on the electromagnetic spectrum.

2.1.3.8. Assist major and field commands, as requested, to conduct staff assistance visits to ensure subordinate units are meeting compliance with national procedures and policy for federal spectrum management.

2.1.3.9. Evaluate and determine the impact of spectrum-dependent systems on current or planned operational use of the electromagnetic spectrum.

2.1.3.10. Provide assistance to Department of the Air Force activities requiring Joint Spectrum Center services like electromagnetic environmental effects engineering and interference resolution.

2.1.3.11. Provide curriculum input and support to the formal technical spectrum courses when requested by the Joint Staff, representing combatant commands to the Air Force Career Functional Manager.

2.1.3.12. In accordance with the Department of the Air Force's role as the Combatant Command Support Agent, within the United States and its possessions, provide spectrum management support to the following: United States Transportation Command, United States Strategic Command, United States Cyber Command, United States Central Command, United States Northern Command, and United States Special Operations Command IAW Department of Defense Directive 5100.03, Support of the Headquarters of Combatant and Subordinate Unified Command, and ACP 190 US SUPP-1(D). **(T-0)**

2.1.3.13. Develop, coordinate and publish a Department of the Air Force Management Internal Control Toolkit checklist in accordance with AFI 90-201, *The Air Force Inspection System,* and respond to comments related to this Department of the Air Force Instruction as required.

2.1.3.14. Conduct an annual spectrum operations workshop, within funding constraints, open to all levels of Department of the Air Force spectrum operations.

2.1.3.15. Review each spectrum supportability risk assessment ensuring each required assessment component has been accurately completed and presented in accordance with DoDI 4650.01. **(T-0)**

2.1.4. Major and Field Command Spectrum Operations Manager. The manager will:

2.1.4.1. Carry out Air Force policy, practices, and procedures for managing the use of the electromagnetic spectrum. **(T-1)**

2.1.4.2. Ensure compliance with the NTIA frequency assignment review program as required by NTIA Manual, Annex F. **(T-0)**

2.1.4.3. Incorporate wartime and contingency electromagnetic spectrum management procedures into the appropriate operational or contingency plan appendices.

2.1.4.4. Provide electromagnetic spectrum guidance to the commands' acquisition, logistics, intelligence, operations, and communications planning staffs. **(T-1)**

2.1.4.5. Manage electromagnetic spectrum use in the concept, planning, deployment, operation, and evaluation phases of commands' supported exercises and operations. **(T-1)**

2.1.4.6. Process and obtain frequency assignments, spectrum certifications, and International Telecommunication Union satellite registration for spectrum-dependent systems in support of operational requirements in accordance with NTIA Manual and DoDI 4650.01. **(T-0)**

2.1.4.7. Provide guidance on the use of the electromagnetic spectrum early in the concept, exploration, demonstration, and validation phases of the acquisition process.

2.1.4.8. Ensure coordination with the appropriate agencies (i.e., Department of Defense area frequency coordinators) is accomplished prior to frequency assignment in accordance with NTIA Manual and ACP 190 US SUPP-1(D). **(T-0)**

2.1.4.9. Review the subordinate unit electromagnetic spectrum management programs.

2.1.4.10. Assist organizations to ensure there is no degradation of friendly systems or operations during command, control, and communications electromagnetic attack training activities.

2.1.4.11. Ensure subordinates have spectrum processes in place to provide operational support and guidance to users.

2.1.4.12. Provide subordinate spectrum operations personnel with necessary training and guidance required to perform their assigned spectrum duties.

2.1.4.13. Ensure subordinate spectrum operations technicians have current SPECTRUM XXI data.

2.1.4.14. Provide assistance to major or field command's Inspector General's inspecting subordinate organizations to ensure critical compliance items are identified and complied with when accomplishing self-inspections and compliance inspections in accordance with AFI 90-201.

2.1.4.15. Ensure all spectrum-dependent systems employed during sponsored exercises to include foreign military equipment, have approved frequency assignments from an appropriate authority in accordance with the NTIA Manual and ACP 190 US SUPP-1(D). **(T-0)**

2.1.4.16. Compile all components of the Federal Aviation Administration exercise package 150 days before exercise start date with foreign platform participation, in accordance with Federal Aviation Administration, *Interim Guide for Coordinating Identification, Friend or Foe (IFF) Missions*. **(T-0)** Exercise packages include: frequency proposals; participants and platforms; spectrum certifications and/or identification friend or foe; interrogator certification or letter of recommendation; area of responsibilities; and exercise flight schedule.

2.1.4.17. Contact appropriate combatant command for theater specific lead-time criteria.

2.1.4.18. Provide AF/A2/6CO with a schedule of planned exercises within United States and its possessions annually to include dates, times and locations of related planning conferences.

2.1.4.19. Provide quality control and updates to frequency records in the Frequency Resource Record System.

2.1.4.20. Assist in the reporting and resolution of interference events involving assigned frequencies in accordance with AFI 17-221.

2.1.4.21. Review each subordinate unit's spectrum supportability risk assessment submissions ensuring each required assessment component has been properly and accurately completed and presented in accordance with DoDI 4650.01. **(T-0)**

2.1.4.22. Provide administrative support to the Department of the Air Force's Department of Defense area frequency coordinator in accordance with ACP 190 US SUPP-1(D). **(T-0)**

2.1.4.23. Be actively involved in communications and information planning and assist in coordinating and obtaining frequency support to meet the commands' mission.

2.1.5. Numbered Air Force Spectrum Operations Manager. The manager will carry out Department of the Air Force policy, practices, and procedures for managing use of the electromagnetic spectrum. **(T-2)** Numbered Air Force Spectrum Operations Manager will:

2.1.5.1. Participate in communications and information planning, assist in coordinating, and obtaining frequency support to meet the major command, air component, and/or combatant command missions.

2.1.5.2. Provide major command spectrum supportability guidance to the planning staff functions that includes the Department of the Air Force spectrum certification and radio frequency assignment processes and requirements.

2.1.5.3. Evaluate and coordinate spectrum use during the concept, planning, deployment, operation, and evaluation phases of exercises and operations.

2.1.5.4. Obtain frequency assignments and spectrum certifications for spectrum-dependent systems in accordance with the NTIA Manual and DoDI 4650.01, in support of installation infrastructures, exercises, contingencies, or wartime operational requirements. **(T-0)**

2.1.5.4.1. Ensure all frequency requirements and allotment requests for operations within the United States and its possessions are coordinated with the appropriate

Department of Defense area frequency coordinator, major command, and/or service spectrum management office with jurisdiction in the proposed area of use. **(T-2)**

2.1.5.4.2. Ensure all frequency requirements and allotment requests for operations outside the United States and its possessions are coordinated with the appropriate combatant command Joint Frequency Management Office in accordance with procedures outlined in Chairman of the Joint Chiefs of Staff Manual (CJCSM) 3320.01C, *Joint Electromagnetic Spectrum Operations in the Electromagnetic Operational Environment,* and combatant command guidance. **(T-0)**

2.1.5.5. Maintain current editions of the frequency management publications.

2.1.5.6. Maintain an accurate SPECTRUM XXI database of all frequency assignments.

2.1.5.7. Ensure frequency assignments are reviewed according to guidance provided in the NTIA Manual, Annex F. **(T-0)**

2.1.6. Major and Field Command, Subordinate Unit Commander. A unit commander assigned to the Department of the Air Force test and training ranges, groups, etc. will:

2.1.6.1. Carry out Department of the Air Force policy, practices, and procedures for managing the use of the electromagnetic spectrum within their area of responsibility. **(T-2)**

2.1.6.2. Assist organizations and users within their control regarding communications and information planning, coordinating, and obtaining frequency support to meet the mission.

2.1.6.3. Ensure contingency electromagnetic spectrum management procedures are placed in operational and contingency plans and appendices as appropriate.

2.1.6.4. Provide electromagnetic spectrum guidance to users in acquisition, logistics, intelligence, operations, and communications planning staffs. **(T-2)**

2.1.6.5. Manage electromagnetic spectrum use in the concept, planning, deployment, operation, and evaluation phases, of supported exercises.

2.1.6.6. Provide electromagnetic spectrum guidance to program managers early in the conceptual, exploration, demonstration, and validation phases of the acquisition process. **(T-3)**

2.1.6.7. Review and update frequency assignment records in accordance with guidance found in NTIA Manual, Annex F. **(T-0)**

2.1.7. Host Installation Commander or Director. The commander or director is responsible for all electromagnetic radiation emanating from their installation and from those outlying activities hosted by the installation in accordance with DoDI 4650.01. **(T-0)** Host installation commanders or directors will:

2.1.7.1. Ensure a viable spectrum operations program for the site and areas under the control of the host installation. **(T-2)**

2.1.7.2. Deconflict all spectrum-dependent systems to resolve potential interference to mission essential electromagnetic equipment in accordance with NTIA Manual and DoDI 4650.01. **(T-0)**

2.1.7.3. Appoint a full-time spectrum operations specialist (i.e., 3D1X4) to organize and carry out the spectrum operations program and notify the parent command, appropriate spectrum operations office and the Department of the Air Force Spectrum Management Office. **(T-2)** **Note**: Air National Guard and Air Force Reserve units may identify a full-time member to serve and fulfill the installation spectrum manager role. The use of spectrum operations specialist personnel for Air National Guard and Air Force Reserve Command units is at the Commander's discretion. Non-spectrum operations specialist personnel must receive installation spectrum manager training from the Headquarters Air National Guard or Headquarters Air Force Reserve Command. **(T-2)**

2.1.7.4. Ensure hosted spectrum-dependent systems comply with the NTIA Manual and MC4EB Frequency Panel spectrum policy and guidance. **(T-0)**

2.1.7.5. Ensure spectrum is considered as part of installation encroachment considerations to identify potential mission sustainment hazards in accordance with AFI 90-2001, *Mission Sustainment.* **(T-3)**

2.1.7.6. Ensure appropriate real property procedures are in place and followed (e.g., documentation, competition, fair market value) in order to execute a real estate agreement in accordance with AFI 32-9003, *Granting Temporary Use of Air Force Real Property*, to support deployment of wireless technologies along with other wireless requirements in compliance with Public Law 104-104, *Telecommunications Act of 1996.* **(T-0)** This includes compliance with the requirement that the installation spectrum manager ensures that an electromagnetic compatibility site analysis is performed and the proposed commercial spectrum-dependent system will not negatively impact Department of the Air Force missions in accordance with the AFI 32-9003. **(T-0)**

2.1.7.7. Ensure compliance with the requirements of the Frequency Assignment Review Program described in NTIA Manual, Annex F. **(T-0)**

2.1.8. Installation Communications Commander or Director. The commander or director will:

2.1.8.1. Ensure the installation spectrum manager has, or is able to attain, the security clearance as indicated on the unit manning document required to perform spectrum operations duties. **(T-2)**

2.1.8.2. Ensure the installation spectrum manager has Secret Internet Protocol Router Network access with SPECTRUM XXI and data exchange capability. **(T-3)**

2.1.8.3. Evaluate the complexity of the electromagnetic environment by conducting threat assessments against mission needs using available spectrum analyzers. **(T-3)**

2.1.8.4. Publish an installation spectrum operations instruction and forward draft copies to the parent command for review before publishing. **(T-2)** The local instruction includes:

2.1.8.4.1. Assessment of local purchase or government purchase card orders for spectrum-dependent systems to determine spectrum supportability prior to the obligation of funds; specifically, if the system is deployed outside the continental United States in accordance with ACP 190 US SUPP-1(D), **Chapter 3**. **(T-0)**

2.1.8.4.2. Review of memoranda of understandings or memoranda of agreements that pertain to the use of spectrum-dependent systems. **(T-3)**

2.1.9. Installation and Garrison Spectrum Manager. The spectrum manager is responsible to the installation commander, through their chain of command, for managing all spectrum use on the sites and areas under the control of the installation commander. The spectrum manager will:

2.1.9.1. Ensure using organizations understand the technical parameters and any imposed operational restrictions of their assigned frequencies. **(T-3)**

2.1.9.2. Maintain current frequency management records of all frequencies assigned to the sites and areas hosted by the installation and provide using organizations with a spectrum authorization (license) for their records. **(T-3)**

2.1.9.3. Meet with all using organizations annually to conduct customer education and discuss current spectrum operations issues. **(T-3)**

2.1.9.4. Process frequency proposals and applications for equipment spectrum certification using the End-to-End Supportability System and SPECTRUM XXI respectively, and ensure submissions are processed through the appropriate command channels. **(T-2)**

2.1.9.5. Provide spectrum operations assistance and guidance to tenant activities. **(T-3)**

2.1.9.6. Support local exercise coordination requirements. **(T-3)**

2.1.9.7. Review operational plans and requirement documents, and obtain frequency support through command channels. **(T-3)**

2.1.9.8. Assist deployable units with identifying spectrum requirements for exercises and contingencies. **(T-3)**

2.1.9.9. Review and validate using organizations frequency assignment requirements, validate existing frequency assignment parameters, and submit modifications, renewal, or deletion actions through command channels in accordance with the NTIA Manual, Annex F. **(T-0)**

2.1.9.10. Update frequency records in the Frequency Resource Record System. **(T-3)**

2.1.9.11. Maintain a current point of contact list for using activities. **(T-3)** Contact list includes: name, unit, e-mail address, and phone number. **(T-3)**

2.1.9.12. Ensure contractor activities using the Department of the Air Force's frequency assignments follow federal and MC4EB Pub 7, *Frequency Resource Record System (FRRS) Standard Frequency Action Format (SFAF),* for electromagnetic spectrum policies and guidelines. **(T-0)**

2.1.9.13. Educate spectrum users on the importance of obtaining spectrum supportability guidance and validation prior to entering into a contractual obligation for all spectrum-dependent systems in accordance with the NTIA Manual and DoDI 4650.01. **(T-0)**

2.1.9.14. Perform emitter surveys, based on resource availability, as required with assistance from the parent command to ensure spectrum-dependent systems are operated in accordance with frequency licenses. **(T-3)**

2.1.9.15. Coordinate new and renewal requirements with appropriate Department of Defense area frequency coordinator in accordance with NTIA Manual and ACP 190 US SUPP-1(D). **(T-0)**

2.1.9.16. Immediately notify appropriate Department of Defense area frequency coordinator and command spectrum manager requesting interference resolution assistance to initiate an investigation to locate and determine the offending source and report the interference in accordance with AFI 17-221. **(T-1)**

2.1.9.17. Coordinate with the safety office and weapons safety officer to support the hazard of electromagnetic radiation to ordnance program and provide frequency assignment records for antenna locations in accordance with AFI 91-208, *Hazards of Electromagnetic Radiation to Ordnance (HERO) Certification and Management.* **(T-1)**

2.1.9.17.1. Provide parametric data for spectrum-dependent systems. **(T-3)**

2.1.9.17.2. Coordinate spectrum-dependent system relocation or parametric data changes with the user and safety office. **(T-3)**

2.1.9.17.3. Coordinate with the installation safety office to identify any hazard of electromagnetic radiation to fuel and the bioenvironmental engineering flight to support health hazard evaluations and inventory accuracy in accordance with AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational and Environmental Health Program*. **(T-1)**

2.1.9.18. Submit proposals for units through command spectrum channels. **(T-2)**

2.1.9.19. Submit proposals for tenant units supporting the host mission to the host's and tenant's commands respectively. **(T-2)**

2.1.9.20. Submit proposals for tenant units not supporting the host mission to the supported unit's parent command with a copy to the host and tenant unit spectrum commands. **(T-2)** See examples below:

2.1.9.20.1. Submit frequency requirements for a Headquarters Air Combat Command maintenance expediter net on a Headquarters Air Mobility Command installation to Headquarters Air Combat Command with a copy to Headquarters Air Mobility Command.

2.1.9.20.2. Submit frequency requirements for a Headquarters Air Force Materiel Command unit in support of Air Combat Command on an Air Mobility Command installation to Headquarters Air Combat Command with a copy to Air Force Materiel Command and Air Mobility Command.

2.1.9.21. Ensure all electromagnetic attack events for test, training, and exercises are processed in accordance with CJCSM 3212.02E, *Performing Electronic Attack in the United States and Canada for Tests, Training, and Exercises* or CJCSM 3212.03A, *Performing Tests, Training, and Exercises Impacting the Global Positioning System (GPS).* **(T-0)**

2.1.10. Joint Base Spectrum Operations. The lead service designates a spectrum manager to provide spectrum support for the common functions. The service spectrum manager ensures the Joint Base Spectrum Manager is aware of their spectrum-dependent systems within the boundaries of the Joint Base. Service requirements are processed through national command channels after approval from the Joint Base Spectrum Manager. The Department of the Air Force Joint Base Spectrum Manager will:

2.1.10.1. Pre-coordinate all temporary and permanent frequency proposals with the installation spectrum manager prior to submission to the national level for processing. **(T-2)**

2.1.10.2. Conduct periodic frequency assignment reviews and coordinate modifications on joint frequency assignments prior to submission to the national level in accordance with NTIA Manual, Annex F. **(T-0)**

2.1.10.3. Obtain spectrum certification for joint spectrum-dependent systems prior to submitting any frequency proposal in accordance with the NTIA Manual, Chapter 10 and DoDI 4650.01. **(T-0)**

2.2. User Organization. User organizations, to include but not limited to acquisition, program, operational, test, and tenant units will:

2.2.1. Obtain spectrum certification prior to purchasing a spectrum-dependent system or entering into any contractual obligations for the use of spectrum-dependent devices in accordance with the NTIA Manual, paragraph 10.1.2 and DoDI 4650.01. **(T-0)**

2.2.2. Obtain a frequency assignment through the installation spectrum manager prior to operation of any spectrum-dependent devices that radiate radio frequency energy per NTIA Manual, Chapter 8. **(T-0)**

2.2.3. Maintain a copy of frequency authorizations received from the installation spectrum manager. **(T-3)**

2.2.4. Ensure spectrum-dependent system operations comply with authorized parameters identified in the frequency assignment notification in accordance with NTIA Manual, Chapter 8. **(T-0)**

2.2.5. Act promptly to notify the installation spectrum manager to report and assist in resolving incidents of interference in accordance with AFI 17-221. **(T-1)**

2.2.6. Use radiation-suppression devices when tuning, testing, or experimenting with any equipment that emits radio frequency. **(T-3)**

2.2.7. Maintain and identify a point-of-contact to the installation spectrum manager for unit frequency matters. **(T-3)**

2.2.8. Document and forward documentation for frequency assignment deletions to the installation or garrison spectrum manager. **(T-3)**

2.2.9. Obtain approval from the installation spectrum manager before modifying any existing spectrum-dependent systems or antennas (i.e., increase power, change antenna height or gain), if outside of the assigned parameters of the frequency authorization in accordance with the NTIA Manual, Chapter 10. **(T-0)**

2.2.10. Assist the installation spectrum manager in reviewing and verifying equipment parameters during mandatory and periodic reviews, in accordance with NTIA Manual, Annex F. **(T-0)**

2.2.11. Contact the installation spectrum manager for interpretation or guidance of any national or military-level spectrum management policy. **(T-3)**

2.2.12. Process spectrum supportability risk assessments in accordance with DoDI 4650.01. **(T-0)**

2.2.13. Submit host nation coordination requests to the installation spectrum manager in accordance with CJCSM 3320.01C. **(T-0)**

2.2.14. Obtain electromagnetic attack authorizations from the Department of the Air Force Spectrum Management Office prior to test, training, or exercises in accordance with CJCSM 3212.02E or CJCSM 3212.03A. **(T-0)**

2.3. The Department of Defense Area Frequency Coordinator. The coordinator is responsible for ensuring successful frequency coordination in the areas that lie within, are adjacent to, or are within radio line-of-sight to any range spectrum-dependent system, including all systems brought to a national test range or other designated complex in accordance with NTIA Manual, Chapter 8. **(T-0)** The Joint Staff, MC4EB Frequency Panel provides spectrum policy and guidance in accordance with DoDI 4650.01 and ACP 190 US SUPP-1(D).

3. SPECTRUM SUPPORTABILITY ACTIONS.

3.1. Spectrum Certification. Spectrum certification is the statutory process whereby national regulatory bodies, NTIA and Federal Communications Commission review telecommunications systems for compliance with national spectrum standards and determine electromagnetic compatibility. This applies to spectrum-dependent and commercial-off-the-shelf systems, unless specifically exempt, in accordance with the NTIA Manual, Chapter 10.

3.1.1. Local purchases of spectrum-dependent systems will be reviewed by the installation spectrum manager to evaluate spectrum supportability prior to the obligation of funds. If the system will be deployed outside the United States, DAF policy is that host nation coordination is required in accordance with the DoDI 4650.01. **(T-0)**

3.1.2. The NTIA, Spectrum Planning Subcommittee, may approve out-of-band requirements with an analysis and justification. Out-of-band operations are operated on a strict non-interference basis. Program offices provide justification for non-compliant systems in accordance with NTIA Manual, Chapter 10. **(T-0)**

3.1.3. To protect receive only systems (e.g. global/weather broadcast receivers) supporting a critical operation, national level spectrum certification is required. The certification application should contain a narrative that describes the criticality of the operation at a specified location. Once spectrum certification is approved, an assignment request is submitted to the NTIA to protect the receive frequency.

3.1.4. Electromagnetic attack systems are exempt from the national spectrum certification process and are supported through the Joint Staff, MC4EB, Equipment Supportability Permanent Working Group. Systems that are operated outside the United States and its possessions are coordinated through the appropriate combatant command in accordance with DoDI 4650.01. **(T-0)**

3.2. Spectrum Supportability Risk Assessment. An assessment must be performed by the Department of the Air Force acquisition programs and materiel developers to identify risk as early as possible and affect design and procurement decisions for all spectrum-dependent systems. Review these risks at each milestone decision point as outlined in DoDI 4650.01. **(T-0)**

3.2.1. Department of the Air Force acquisition or materiel development programs, including electromagnetic attack systems, will submit spectrum supportability risk assessment at each milestone decision point in accordance with DoDI 4650.01. **(T-0)**

3.2.2. Quick response programs enter at Milestone C (production and deployment) and go directly to initial operational test and evaluation in accordance with DoDI 4650.01. **(T-0)**

3.2.3. The program manager will insert the NTIA certification stage and number, and approval date into the Information Technology Investment Portfolio System in accordance with AFI 17-110, *Information Technology Portfolio Management and Capital Planning and Investment Control*.

3.3. Department of Defense Electromagnetic Compatibility Standards. Military-Standard-464, *Electromagnetic Environmental Effects (E3) Requirements for Systems,* provides important electromagnetic compatibility and electromagnetic environmental effects standards for Department of Defense spectrum-dependent systems. The Department of the Air Force spectrum programs will adhere to Military-Standard-464 and DoDI 3222.03, *DoD Electromagnetic Environmental Effects (E3) Program*. **(T-0)**

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

Public Law 104-104, *Telecommunications Act of 1996*, 8 February 1996

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AFI 17-221, *Spectrum Interference Resolution Program*, 11 May 2018

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Prescribed Forms

None

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

MC4EB Pub 7, *Frequency Resource Record System (FRRS) Standard Frequency Action Format (SFAF)*

Abbreviations and Acronyms

ACP—Allied Communications Publication

AFI—Air Force Instruction

CJCSM—Chairman of the Joint Chiefs of Staff Manual

DoDI—Department of Defense Instruction

E3—Electromagnetic Environmental Effects

ITU RR—International Telecommunication Union Radio Regulation

JP—Joint Publication

MC4EB—Military Command, Control, Communications, and Computers Executive Board

NTIA—National Telecommunications and Information Administration

Terms

Allotment (of a radio frequency or radio frequency channel)—Entry of a designated frequency channel in an agreed plan, adopted by a component conference, for use by one or more administrations for a (terrestrial or space) radio communication service in one or more identified countries or geographical areas and under specified conditions. Source: ITU RR.

Assignment (of a radio frequency or radio frequency channel)—Authorization given by an administration for a radio station to use a radio frequency or radio frequency channel under specified conditions. Source: ITU RR.

Commercial—off-the-Shelf—Spectrum-dependent systems that can be procured by the general public wholesale or retail. Source: National Institute of Standards and Technology.

Electromagnetic Attack—Division of electromagnetic warfare involving the use of electromagnetic energy, directed energy, or anti-radiation weapons to attack personnel, facilities, or equipment with the intent of degrading, neutralizing, or destroying enemy combat capability and is considered a form of fires. Source: JP 3-85.

Electromagnetic Compatibility—(1) The condition that prevails when telecommunications equipment is performing its individually designed function in a common electromagnetic without causing or suffering unacceptable degradation due to unintentional electromagnetic interference to or from other equipment in the same environment. Source: NTIA. (2) The ability of systems, equipment, and devices that use the electromagnetic spectrum to operate in their intended environments without causing or suffering unacceptable or unintentional degradation because of electromagnetic radiation or response. Source: JP 3-85

Electromagnetic Environmental Effects (E3)—The impact of the electromagnetic operational environment upon the operational capability of military forces, equipment, systems, and platforms. E3 encompasses the electromagnetic effects addressed by the disciplines of electromagnetic compatibility, electromagnetic interference, electromagnetic vulnerability, electromagnetic pulse, electromagnetic protection, electrostatic discharge, and electromagnetic radiation hazards to personnel, ordnance, and fuels and volatile materials. E3 includes the affects generated by all electromagnetic environment contributors including radio frequency systems, ultra-wideband devices, high-powered microwave systems, lightning, and precipitation static. Source: DoDI 3222.03.

Electromagnetic Spectrum Management—The operational, engineering, and administrative procedures to plan, and coordinate operations within the electromagnetic operational environment. Source: JP 3-85.

Electromagnetic Spectrum Operations—Coordinated military actions to exploit, attack, protect, and manage the electromagnetic environment. Source: JP 3-85.

Electromagnetic Warfare—Military action involving the use of the electromagnetic and directed energy to control the electromagnetic spectrum or to attack the enemy. Source: JP 3-85.

Frequency Assignment—see Assignment (of a radio frequency or radio frequency channel). Source: ITU RR.

Frequency Coordination—(1) The process of obtaining approval to use the radio frequency spectrum via arrangements and technical liaison for the purpose of minimizing harmful interference through cooperative use of the radio frequency spectrum. Source: NTIA. (2) Rules and mechanisms that control how to use the electromagnetic spectrum in specified dimensions (i.e., spatial, time, frequency, power, waveform. Source: JP 3-85. To be effective, the coordination must extend through the planning, proposal, and actual in use phases of radio frequency utilization.

Interference—(1) The effect of unwanted energy due to one or a combination of emissions, radiations, or inductions upon reception in a radio communication system, manifested by any performance degradation, misinterpretation, or loss of information that could be extracted in the absence of such unwanted energy. Source: ITU RR. (2) Any electromagnetic disturbance, induced intentionally or unintentionally, that interrupts, obstructs, or otherwise degrades or limits the effective performance of electromagnetic spectrum-dependent systems and electrical equipment. Also referred to as electromagnetic interference. Source: JP 3-85.

Radio Frequency Spectrum—The radio frequency spectrum includes the frequencies from 3.0 kilohertz to 400 gigahertz. The presently allocated spectrum is from 9 kilohertz to 275 gigahertz. Also called electromagnetic spectrum. Source: NTIA.

Spectrum Supportability Risk Assessment—An assessment performed by Department of the Defense for all spectrum-dependent systems to identify risk as early as possible and affect design and procurement decisions. These risks are reviewed at acquisition milestones and managed throughout the system's life cycle. Source: DoDI 4650.01.

National Table of Allocations—Entry in the federal table of frequency allocations of a given frequency band for its use by one or more (terrestrial or space) radio communication services or the radio astronomy service under specified conditions. This term also applies to the frequency band concerned. Source: NTIA.

Telecommunication—Any transmission, emission, or reception of signs, signals, writings, images, and sounds or information of any nature by wire, radio, visual or other electromagnetic systems. Source: ITA RR.

United States and its Possessions—Includes the 50 States, District of Columbia, the Commonwealth of Puerto Rico, and the territories and possessions (but less the Canal Zone) Source: NTIA.